

AN ORDER-128 2-GROUP WHOSE SQUARES FORM A NON-POWERFUL SUBGROUP

ABSTRACT. Problem 21.137 of the Kourovka Notebook, proposed by L. E. Wilson, asks: if the p -th powers in a finite p -group form a subgroup, must that subgroup be powerful? Its $p = 2$ form asks whether, in a 2-group of exponent 8 whose squares form a subgroup, that subgroup must be abelian. We answer both in the negative with $G = \text{Syl}_2(S_8) = (C_2 \wr C_2) \wr C_2$ of order $2^7 = 128$. The set $S = \{g^2 : g \in G\}$ is a subgroup of order 16, isomorphic to $C_2 \times D_8$ and hence non-abelian; since $\exp S = 4$ forces $S^4 = 1$ while $|S'| = 2$, S is not powerful in the published convention $H' \leq P_2(H) = \langle h^4 \rangle$ for $p = 2$. The group is exhibited both as seven explicit permutations of $\{1, \dots, 8\}$ and by a 28-relator presentation. The odd-prime question in the second sentence of 21.137 is a different question and is not addressed here.

1. THE QUESTION

The Kourovka Notebook [1] records, as Problem 21.137 with proposer L. E. Wilson, the following. All three of its sentences are quoted verbatim.

21.137. If the p -th powers in a finite p -group form a subgroup, must that subgroup be powerful? That is, for $p \neq 2$, if the p -th powers in a p -group of exponent p^2 form a subgroup, must that subgroup be abelian? For a 2-group of exponent 8, if the squares form a subgroup, must that subgroup be abelian? — *L. Wilson*

Here “the p -th powers form a subgroup” means that the set $\{g^p : g \in G\}$ is closed under multiplication, so that it coincides with the power subgroup $P_1(G) = \langle g^p : g \in G \rangle$ it generates. One direction of the implication 21.137 asks about is a theorem: for odd p , in a powerful p -group the set of p^k -th powers is the subgroup it generates (Lubotzky and Mann [2] for $k = 1$, and the proposer [3, Thm. 3.1] for every k). Problem 21.137 asks for the converse.

The one definition the problem does not restate is the one that decides it. Following Lubotzky and Mann [2, §1], a finite p -group H is *powerful* when

$$(1) \quad \begin{aligned} H' &\leq P_1(H) = \langle h^p : h \in H \rangle \quad \text{for odd } p, \\ H' &\leq P_2(H) = \langle h^4 : h \in H \rangle \quad \text{for } p = 2. \end{aligned}$$

2020 *Mathematics Subject Classification.* 20D15, 20E22, 20F05.

Key words and phrases. powerful p -group, power subgroup, set of squares, Kourovka Notebook, Sylow subgroup of the symmetric group, counterexample.

The shift to *fourth* powers at $p = 2$ is not a convenience: with $\langle h^2 \rangle$ in place of $\langle h^4 \rangle$ the condition would hold in every group whatsoever (Lemma 3 below), so the $p = 2$ half of the headline question would be trivially affirmative.

Convention (1) is what makes the problem's own bridge sentence ("That is, ...") correct at $p = 2$ as well as at odd p . If $\exp G = 8$ and $S = \{g^2 : g \in G\}$ is a subgroup, then $\exp S = 4$, so $S^4 := P_2(S) = 1$ and (1) reduces to $S' = 1$. Thus for $p = 2$, "powerful" and "abelian" say the same thing about S , and the third sentence of 21.137 is a faithful specialisation of the first, not a weaker or a stronger question.

Theorem 1. *Let $G = \text{Syl}_2(S_8) = (C_2 \wr C_2) \wr C_2$, a group of order $2^7 = 128$, and let $S = \{g^2 : g \in G\}$. Then*

- (i) $\exp G = 8$;
- (ii) S is a subgroup of G , of order 16, and $S = P_1(G) = G' = \Phi(G)$;
- (iii) $S \cong C_2 \times D_8$; in particular S is **not abelian**, and $|S'| = 2$;
- (iv) $\exp S = 4$, so $P_2(S) = 1$, and therefore $S' \not\leq P_2(S)$: the subgroup S is **not powerful**.

Consequently the answer to the headline question of 21.137 is no, already at $p = 2$, and the answer to its third sentence is no.

2. THE OBJECT

Write $[x, y] = x^{-1}y^{-1}xy$, so that $xy = yx[x, y]$. Let $G \leq S_8$ be generated by the seven permutations

$$\begin{array}{ll} f_1 = (1\ 5)(2\ 6)(3\ 7)(4\ 8) & f_5 = (1\ 3)(2\ 4)(5\ 7)(6\ 8) \\ f_2 = (3\ 4) & f_6 = (5\ 6)(7\ 8) \\ f_3 = (5\ 7)(6\ 8) & f_7 = (1\ 2)(3\ 4)(5\ 6)(7\ 8) \\ f_4 = (3\ 4)(7\ 8) & \end{array}$$

These seven satisfy, and are presented by, the 28 relators

$$\begin{aligned} f_1^2 = f_2^2 = f_3^2 = f_4^2 = f_5^2 = f_6^2 = f_7^2 = 1, \\ [f_2, f_1] = f_4, \quad [f_3, f_1] = f_5, \quad [f_6, f_1] = f_7, \\ [f_5, f_2] = f_6f_7, \quad [f_4, f_3] = f_6, \quad [f_5, f_4] = f_7, \\ \text{and } [f_a, f_b] = 1 \text{ for each of the remaining 15 pairs } a > b. \end{aligned} \tag{2}$$

That is seven squares together with all $\binom{7}{2} = 21$ commutators, six of them non-trivial. Every right-hand side in (2) is a word in generators of index strictly larger than both entries of its commutator, so (2) is a polycyclic presentation with all seven relative orders equal to 2: collecting an arbitrary word to the left terminates, and the presented group is spanned by the 128 normal words

$$f_1^{a_1} f_2^{a_2} \cdots f_7^{a_7}, \quad a_i \in \{0, 1\}. \tag{3}$$

In G those 128 words are pairwise distinct, and $|G| = 128$ by Lemma 2(b); hence (2) presents G exactly, the words (3) are its elements. Two further remarks fix its identity. First, $G \leq S_8$ and $8! = 40320 = 128 \cdot 315$ with 315 odd,

so a subgroup of S_8 of order 128 is a full Sylow 2-subgroup: $G \in \text{Syl}_2(S_8)$. Second, G is also the closure of the seven level generators $(1\ 2)$, $(3\ 4)$, $(5\ 6)$, $(7\ 8)$, $(1\ 3)(2\ 4)$, $(5\ 7)(6\ 8)$, $(1\ 5)(2\ 6)(3\ 7)(4\ 8)$ of the automorphism group of the complete binary tree with leaves $1, \dots, 8$, which is $(C_2 \wr C_2) \wr C_2$.

Put

$$N = \langle f_4, f_5, f_6, f_7 \rangle.$$

Lemma 2. (a) $N \cong D_8 \times C_2$: it has order 16, $N' = \langle f_7 \rangle$ has order 2, and $\exp N = 4$.

(b) $N \trianglelefteq G$ and $G/N \cong C_2^3$; in particular $|G| = 128$ and $g^2 \in N$ for every $g \in G$.

(c) Every element of N is a square in G . Hence $S = \{g^2 : g \in G\} = N$.

(d) $\exp G = 8$.

Proof. (a) Among the six pairs drawn from f_4, f_5, f_6, f_7 , the only non-trivial commutator in (2) is $[f_5, f_4] = f_7$. So f_6 and f_7 are central in N . As all four generators are involutions, $[f_5, f_4] = f_5 f_4 f_5 f_4 = (f_5 f_4)^2$, whence $(f_5 f_4)^2 = f_7$ and, taking inverses, $r := f_4 f_5$ satisfies $r^2 = f_7$ and $r^4 = f_7^2 = 1$. Since $f_7 \neq 1$, r has order exactly 4; and $f_4 r f_4 = f_4 f_4 f_5 f_4 = f_5 f_4 = r^{-1}$ with $f_4 \notin \langle r \rangle = \{1, r, f_7, r f_7\}$, so $\langle f_4, f_5 \rangle = \langle r, f_4 \rangle \cong D_8$ of order 8, with derived subgroup $\langle f_7 \rangle$. Finally f_6 is central in N and, by uniqueness of the normal form (3), $f_6 \notin \langle f_4, f_5 \rangle$; so $N = \langle f_4, f_5 \rangle \times \langle f_6 \rangle \cong D_8 \times C_2$, of order 16 and exponent 4, with $N' = \langle f_7 \rangle$ of order 2.

(b) Each of $[f_4, f_i], [f_5, f_i], [f_6, f_i], [f_7, f_i]$ for $i = 1, 2, 3$ lies in N : by (2) the only non-trivial ones are $[f_6, f_1] = f_7$, $[f_5, f_2] = f_6 f_7$ and $[f_4, f_3] = f_6$. As N is generated by $f_4, \dots, f_7$ and G by $f_1, \dots, f_7$, this gives $N \trianglelefteq G$. Modulo N the images of f_1, f_2, f_3 are involutions which commute, because $[f_2, f_1] = f_4 \in N$, $[f_3, f_1] = f_5 \in N$ and $[f_3, f_2] = 1$; so G/N is generated by three commuting involutions, and it has order 8 because the 128 normal words (3) are distinct in G while $|N| = 16$. Thus $G/N \cong C_2^3$ and $|G| = 16 \cdot 8 = 128$. Since G/N has exponent 2, $(gN)^2 = N$, i.e. $g^2 \in N$, for every $g \in G$.

(c) Table 1 lists, for each of the sixteen elements s of N , an element $r \in G$ with $r^2 = s$. Two entries are immediate by hand: from $[f_2, f_1] = f_4$ and $f_1^2 = f_2^2 = 1$ we get $(f_2 f_1)^2 = f_2 f_1 f_2 f_1 = [f_2, f_1] = f_4$, hence $(f_1 f_2)^2 = ((f_2 f_1)^2)^{-1} = f_4$, and likewise $(f_1 f_3)^2 = f_5$. Together with (b), $S = N$.

(d) By (a) and (c), $S = N$ has exponent 4, so $g^8 = (g^2)^4 = 1$ for every $g \in G$ and $\exp G$ divides 8. Conversely $f_4 f_5 \in N$ has order 4 by (a) and is a square by (c) — Table 1 gives $(f_1 f_2 f_3 f_5)^2 = f_4 f_5$ — so $f_1 f_2 f_3 f_5$ has order 8. Hence $\exp G = 8$ exactly. $\square$

Proof of Theorem 1. Part (i) is Lemma 2(d). By Lemma 2(b,c), $S = N$ is a subgroup of order 16, and it equals $P_1(G) = \langle g^2 : g \in G \rangle$ because the set of squares already is a subgroup. For $S = G'$: by Lemma 2(b) the quotient G/N is abelian, so $G' \leq N$, while (2) exhibits all four generators of N as commutators of elements of G , namely $f_4 = [f_2, f_1]$, $f_5 = [f_3, f_1]$, $f_6 = [f_4, f_3]$ and $f_7 = [f_6, f_1]$, so $N \leq G'$; hence $G' = N = S$. Since

TABLE 1. The sixteen elements s of $S = N$, each with a square root $r \in G$ and its order. Words are read left to right in $f_1, \dots, f_7$.

s	r with $r^2 = s$	$ r $	s	r with $r^2 = s$	$ r $
1	1	1	f_5f_6	$f_1f_3f_4$	4
f_4	f_1f_2	4	f_5f_7	$f_1f_3f_6$	4
f_5	f_1f_3	4	f_6f_7	f_2f_5	4
f_6	f_3f_4	4	$f_4f_5f_6$	$f_1f_2f_3f_6$	8
f_7	f_1f_6	4	$f_4f_5f_7$	$f_1f_2f_3f_4$	8
f_4f_5	$f_1f_2f_3f_5$	8	$f_4f_6f_7$	$f_1f_2f_5$	4
f_4f_6	$f_1f_2f_4f_5$	4	$f_5f_6f_7$	$f_1f_3f_4f_5$	4
f_4f_7	$f_1f_2f_6$	4	$f_4f_5f_6f_7$	$f_1f_2f_3$	8

$\Phi(G) = G'P_1(G)$ for a p -group, $\Phi(G) = S$ as well, and (ii) holds. Part (iii) is Lemma 2(a); explicitly, $f_4 = (f_1f_2)^2$ and $f_5 = (f_1f_3)^2$ are two squares with

$$[f_4, f_5] = f_7 \neq 1,$$

so S is not abelian, and $S' = \langle f_7 \rangle$ has order 2. For (iv), $\exp S = 4$ gives $s^4 = 1$ for every $s \in S$, hence $P_2(S) = \langle s^4 \rangle = 1$, while $S' \neq 1$; therefore $S' \not\leq P_2(S)$ and S is not powerful by (1). Since G is a finite 2-group whose squares do form a subgroup, the hypotheses of both the first and the third sentence of 21.137 are met and both are answered in the negative. $\square$

Lemma 3. *In every group H one has $H' \leq \langle h^2 : h \in H \rangle$.*

Proof. Put $K = \langle h^2 : h \in H \rangle$. Then $K \trianglelefteq H$, and every element of H/K squares to the identity, so H/K has exponent at most 2. A group of exponent at most 2 is abelian, since $xy = (xy)^{-1} = y^{-1}x^{-1} = yx$. Hence H/K is abelian and $H' \leq K$. $\square$

3. SCOPE

Only the headline sentence of 21.137 and its third sentence are answered above. The *second* sentence, “for $p \neq 2$, if the p -th powers in a p -group of exponent p^2 form a subgroup, must that subgroup be abelian?”, is a different question, no object in this note bears on it, and it is not addressed here. We do not claim that 128 is the least order at which the phenomenon occurs, nor that G is the only witness of its order.

We make no priority claim. We have not found Theorem 1 in print, but our search of the literature was not exhaustive and $\text{Syl}_2(S_8)$ is an obvious group to test, so an exercise-level or folklore appearance elsewhere is possible.

4. REPRODUCTION AND VERIFICATION

Everything above can be redone from the printed data alone: enter the seven permutations into any computer algebra system, close under multiplication to obtain 128 elements, form the set of squares and test it for closure. Lemma 2 and the proof of Theorem 1 use nothing beyond (2) and Table 1 and can be checked with a pencil.

The accompanying program `verify.py` re-derives the quantities asserted here in exact integer and permutation arithmetic, using the Python standard library only and no group-theory package; `verify.output.txt` records a run of it. Its input is the printed data: the seven permutations, the 28 relators (2) and the sixteen roots of Table 1. From these it verifies the relators, rebuilds G by closure and confirms $|G| = 128$ and $\exp G = 8$; checks that the 128 normal words (3) are distinct and that collection by the printed relators rewrites all $128^2 = 16384$ products of normal words correctly, which bounds the presented group by 128 and so makes (2) a presentation of G ; tests all 256 ordered products in S for closure, obtaining $|S| = 16$ and $S = N = G' = \Phi(G) = P_1(G)$; confirms the sixteen square roots of Table 1 and their orders; and computes $\exp S = 4$, $P_2(S) = 1$ and $|S'| = 2$, whence $S' \not\leq P_2(S)$.

REFERENCES

- [1] E. I. Khukhro and V. D. Mazurov (eds.), *Unsolved Problems in Group Theory. The Kourovka Notebook. No. 21*, Sobolev Institute of Mathematics, Novosibirsk. Cited from the L^AT_EX e-print arXiv:1401.0300v45; Problem 21.137, proposed by L. E. Wilson.
- [2] A. Lubotzky and A. Mann, *Powerful p -groups. I. Finite groups*, J. Algebra **105** (1987), no. 2, 484–505. doi:10.1016/0021-8693(87)90211-0. The source of (1), including the clause that replaces p -th powers by fourth powers when $p = 2$.
- [3] L. E. Wilson, *On the power structure of powerful p -groups*, J. Group Theory **5** (2002), no. 2, 129–144. doi:10.1515/jgth.5.2.129. Proves, for odd p , that in a powerful p -group the set of p^k -th powers is a subgroup; 21.137 asks for the converse.